

**PURCHASE DESCRIPTION FOR THE RETROFIT AND
UPGRADE OF AN AJAX MOLTEN SALT BATH HEAT-
TREATING TANK FURNACE AND QUENCH TANK
REPLACEMENT**

1.0 SCOPE.

1.1 Scope. The content of this specification describes the minimum requirements for the upgrade of the existing Ajax-Hultgren Salt Bath Furnace. The Contractor shall provide all material and labor for a turnkey upgrade to the existing salt bath heat-treating tank furnace system located at Fleet Readiness Southwest, Naval Air Station North Island, San Diego California, Building 472. The Government shall be responsible for providing the “core” Molten Salt Bath Tank Furnace to upgrade and water quench tank. The furnace to be upgraded is an Ajax molten salt bath furnace #10, serial# 4217-A and PEMA# 65888007617. The Contractor shall upgrade the furnace in accordance with the requirements specified herein.

2.0 APPLICABLE DOCUMENTS. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed in section 2. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

2.1 Government Documents.

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein.

STANDARDS – FEDERAL

FED-STD-H28B: Screw Thread Standards for Federal Service

**U.S Department of Labor, Occupational Safety and Health Administration
(OSHA)**

29 CFR 1910: General Industry, OSHA Safety and Health Standards

(Application for copies should be addressed to the OSHA Publications
Superintendent of Documents P.O. Box 37535 Washington, D.C. 20013-7535)

Environmental Protection Agency (EPA)

40 CFR 761: Polychlorinated Biphenyls (PCBs) Manufacturing, Processing,
Distribution in Commerce, and Use Prohibitions

(Application for copies should be addressed to the EPA Ariel Rios Building
1200 Pennsylvania Avenue, N.W. Washington, DC 20460)

DEPARTMENT OF DEFENSE

U.S. NAVY

OPNAV 5100.23H: Navy Safety and Occupational Health Program Manual

(Application for copies of the OPNAVINST should be addressed to Office of the
Chief of Naval Operations, The Pentagon, Washington, DC 20350-2000 or to
Commander, Naval Safety Center 375 A Street Norfolk, VA 23511 or on the
internet at <http://www.public.navy.mil/navsafecen>)

MILITARY STANDARDS

MIL-STD-130N w/Change 1: Identification Marking of U.S. Military Property
Item Unique Identification Marking (IUID)

MIL-S-10699: Salt, Heat Treating for Metals

(Application for copies should be addressed to Air Force Product Data Systems
Modernization Standards (TMSS) Office, 554 ELSG/SBP, 4375 Chidlaw Road,
Bldg. 262, Room S008, Wright-Patterson AFB, OH 45433-5006, Phone (937) 257-
0870 or 0853 or <http://quicksearch.dla.mil/qsSearch.aspx>)

2.1.2 Other Government Documents, Drawings, and Publications. The following other
Government documents, drawings, and publications form a part of this document to the
extent specified herein. Unless otherwise specified, the documents listed in the solicitation
or contract shall be the current edition at time of solicitation.

LOCAL FACILITY DOCUMENTS

RESERVED

2.2 Non-Government Publications, Specifications, Standards, and Handbooks. The
following documents form a part of this document to the extent specified herein. Unless
otherwise specified, the documents listed in the solicitation or contract shall be the current
edition at time of solicitation.

American Society of Mechanical Engineers (ASME)

ASME Y14.100: Engineering Drawing Practices

(Applications for copies should be addressed to ASME International Three Park
Avenue New York, NY 10016-5990 or www.asme.org)

(Address application for copies to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036)

International Organization for Standardization (ISO)

ISO 4414: Pneumatic Power-General rules and safety requirements for systems and their components

(Application for copies should be addressed to the International Organization for Standardization (ISO) 1 rue de Varembé, Case postale 56, CH-2011 Geneva 20, Switzerland)

National Electrical Manufacturers Association (NEMA)

NEMA 250: Enclosures for Electrical Equipment (1000 V Maximum)

NEMA ICS 1: Industrial Control and Systems General Requirements (2010 E)

NEMA ICS 6: Industrial Control and Systems: Enclosures (2011 R)

NEMA MG 1: Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 N 17th Street, Rosslyn, VA 22209)

National Fire Protection Association (NFPA)

NFPA 70: National Electrical Code

NFPA 79: Electrical Standard for Industrial Machinery

NFPA 86: Standard for Ovens and Furnaces

(Applications for copies should be addressed to the National Fire Protection Association, ATTN: Support Services, 11 Tracy Drive, Avon, MA 02322)

American Welding Society (AWS)

AWS D1.1/D1.1M:2020 Structural Welding Code – Steel, American Welding Society

(Applications for copies should be addressed to the American Welding Society, 8669 NW 36 Street, #130 Miami, Florida 33166-6672 or <https://www.aws.org/>)

Society of Automotive Engineers (SAE)

SAE AEROSPACE MATERIAL SPECIFICATION (AMS) 2750E: I Pyrometry

SAE AMS 2821: Salt, Heat Treating for Metals

(Application for copies should be addressed to: Society of Automotive Engineers
400 Commonwealth Drive, Warrendale PA 15096)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

3.0 REQUIREMENTS.

3.1 Expectation. The Contractor shall be responsible for furnishing all services, qualified personnel, parts and materials necessary to upgrade the existing furnace in accordance with all requirements specified herein. Upgraded parts shall be installed such that normal maintenance, adjustment, replacement, and inspection can be performed safely and efficiently.

3.1.1 Measuring and Indicating Device Graduations. All measuring and indicating devices on the equipment shall be graduated in the United States customary units. An electronic control system that demonstrates switching between manufacturer standard units to United States customary units will be found acceptable.

3.1.2 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's workstation(s). The control system, including safety interlocks and shutdown functions, shall comply with requirements of NFPA 86.

3.1.3 Safety and Health Requirements. All equipment parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 or equivalent ISO/EN/DIN/JIS standards. At the time of the proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.1.4 Audible Noise Level. Audible noise emitted by the equipment shall not exceed 84 decibels (dB), measured on the "A" weighted scale of a standard Type II sound level meter, at the operator's work position or any point at a distance of three (3) feet from the equipment. Noise generated by the work piece shall be excluded in determining compliance of the equipment with the 84 dB requirements.

3.1.5 Hazardous Material Exclusions. Supplies or materials being provided as part of the equipment shall be free of known hazardous materials. These materials shall not be brought on site without prior approval from the Government. Radioactive materials or instruments capable of producing ionizing radiation as well as materials which contain asbestos, mercury, lithium, methylene chloride, lead ($\geq 0.06\%$), or polychlorinated biphenyls (PCB's) are prohibited. Definitions of hazardous materials are specified in the latest version, including revisions adopted during the term of the contract, of FED-STD-313F. Nickel Metal Hydride and lithium batteries are permissible for memory backup. Class I Ozone Depleting Substances as defined in 40 CFR 82 shall not be used in the performance

of this contract or be provided as part of the equipment. Exceptions to the prohibition of these materials must be referred to the Contracting Officer in writing, for consideration, not later than 60 calendar days after effective date of contract and prior to any work being performed.

3.1.5.1 Use of Polychlorinated Biphenyl (PCB). The use of polychlorinated biphenyl (PCB) on or in the equipment is prohibited. The equipment shall be considered PCB free providing certification to be verified containing less than 5 parts per million (<5ppm) in accordance with 40 CFR 761.

3.1.6 Environmental Protection. The completed upgrade shall meet all applicable Environmental Protection Agency (EPA) and Occupational Safety Health Association (OSHA) restriction for materials classified as hazardous to the environment in effect on the date of contract. During the operation of the equipment at final location, all equipment shall comply with all applicable EPA and OSHA regulations in effect on the date of contract regulating the emission of material hazardous to the environment. The contractor shall coordinate with FRCSW and Naval Air Station, North Island (NAS) regarding installation site approvals and environmental permits required by Federal, State, County, and Local Laws. For additional information, the contractor is instructed to contact the Government representative. Within thirty (30) days from contract award, contractor shall provide in writing the make, model, capacity, footprint, mechanical, and electrical requirements of the equipment to the Government Representative.

3.1.7 Replacement Parts. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Replacement parts are those that are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall either be interchangeable with domestically supplied components or be readily available from a domestic source.

3.2 Construction. The machine shall be constructed of parts that are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.2.1 Castings and Forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 Fastening Devices. All screws, pins, bolts, and other fasteners shall be installed to prevent unintentional loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.2.3 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28B and the applicable "Detailed Standard" section referenced therein or equivalent ISO/EN/DIN/JIS standards. At the time of

proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.2.4 Pneumatic System. Under the condition that a machine has a pneumatic system, the system shall conform to ISO 4414. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls.

3.2.5 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. The condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.6 Welding, Brazing, or Soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.2.7 Painting. The machine shall be painted in accordance with the manufacturer's commercial practice. All paint shall be of the lead-free type.

3.2.8 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this purchase description.

3.3 Components. The machine shall consist of not less than the following components:

3.3.1 Molten Salt Bath Upgrade. The Contractor shall provide and install a new natural gas (NG) heating source system that shall be capable of heating and controlling the molten salt bath to an operating temperature range of 850°F to 990°F, with $\pm 5^\circ\text{F}$ control stability across the molten salt bath under steady-state conditions. Upon completion of the upgrade, the furnace shall be capable of reaching a maximum operating temperature of 1100°F and shall be capable of meeting Class 1 $\pm 5.0^\circ\text{F}$ furnace performance requirements IAW SAE AMS 2750.

3.3.1.1 Salt Bath Tank and Liner. Contractor shall remove and replace existing 66" wide X 153" long X 112" deep tank liner and associated brick insulation. New refractory brick-type insulation shall be installed throughout the entire new liner and existing secondary tank containment in accordance with the furnace manufacturer's written installation requirements, NFPA 86, and SAE AMS 2750 requirements for furnace performance. Any welded structural elements shall comply with AWS D1.1/D1.1M.

3.3.1.2 Agitation System. The Contractor shall provide a new impeller agitation system consisting of 4 each individual units, ensuring proper agitation and temperature uniformity throughout the molten salt bath.

3.3.1.3 Furnace Lid and Rail System. The Contractor shall replace the existing salt bath furnace lid, rail system, and assisted pneumatic open/close mechanism to accommodate new agitation system. This door design must minimize the heat loss of the furnace to pass the TUS IAW SAE AMS 2750.

3.3.2 Quench Tank. Contractor shall provide a new matching ambient temperature water quench tank. The tank shall be fabricated from stainless steel and must have a volume of 66" wide X 153" long X 112" deep. The tank shall be provided with an enclosed side agitation system.

3.3.3 Solution. Contractor shall furnish and install a new heat-treating salt solution consisting of 50% sodium nitrate and 50% potassium nitrate, in sufficient quantity to completely charge the salt bath to the manufacturer-recommended operating level. The solution shall conform with a Class 2 heat treating salt solution, per MIL-S-10699 and SAE AMS 2821.

3.3.4 Natural Gas Burner System Capacity and Safety. The Contractor shall provide and install two (2) natural gas radiant burner heating elements with a combined total heat input rating of 1,500,000 BTU (or as otherwise demonstrated necessary to meet the performance requirements of this SOW). The heating system shall include all required safety features IAW NFPA 86, including flame safety system, over-temperature protection system, automatic ignition system, and safe shutdown on fault or loss of flame.

3.3.5 Electrical System. Contractor shall be responsible for the removal and replacement of the existing electrical cabinet. The cabinet shall conform to NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of the proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. The existing source power available at the facility is 480 volt, 3-phase, 60 Hertz (Hz), 250KVA total output. The equipment's electrical system shall be tolerant of fluctuation of ± 10 percent. The electrical system shall be complete, including any electrical transformer(s) that may be required to modify the existing source voltage to the proper operating voltage of the equipment. A properly rated, and fused, single disconnect device shall be utilized on the machine with means of lockout in the "off" position only.

3.3.5.1 Electrical Enclosures. Construction of the electrical enclosures shall be in accordance with NEMA 250 type 12 or current equivalent ISO/EN/DIN/JIS/IEC standards and shall provide for ventilation of components and shall be designed for an indoor non-hazardous location. All electrical components shall conform to NFPA 79 or equivalent ISO/EN/DIN/JIS standards. The machine shall be provided with interlocks on its electronic cabinet that shall instantly remove power from the equipment when the door is opened in accordance with NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of the proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.5.2 Electrical Disconnects and Lockout/Tagout Provisions. Each major component of the system shall be provided with a dedicated main power disconnecting means, fused as required and equipped with lockout/tagout (LOTO) capability. All equipment shall be new and shall conform to the latest applicable industry standards and practices. Disconnect switches shall be properly rated for the voltage, amperage, horsepower, and inductive characteristics of the connected load. Switch mechanisms shall be heavy-duty, quick-make, quick-break, and shall be provided with a defeatable door interlock to prevent opening of the enclosure while the switch is in the ON position. Each disconnect shall include an

externally operated, lockable handle with provisions for a minimum of two padlocks. Poles and fusing shall be sized IAW the load requirements. Fused disconnect switches shall comply with NFPA 70 and NFPA 86 requirements.

3.3.5.3 Electrical Power and Control Wiring. The contractor shall provide and install all electrical conduit and power wiring to the main furnace power panel. Contractor shall provide and install all control wiring, switches, and devices necessary for operation of the furnace system. If required, an additional NEMA 4 rated control panel shall be furnished and installed as part of the furnace electrical system.

3.3.5.4 Arc Flash Protection. The machine enclosures shall be marked for arc flash in accordance with NFPA 79 Paragraph 16.2.3 “Safety Signs for Electrical Enclosures” or current equivalent ISO/EN/DIN/JIS/IEC standards for arc flash safety. At the time of the proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.5.5 Solid-State Components. Electrical devices used to power electrical servo and spindle motors of control voltage delivered to solid-state electrical components shall be solid-state devices. Such devices shall be equipped to handle the electrical load imposed by operation of the machine to its maximum potential. The use of selenium in solid-state components shall be restricted to those devices that protect other solid-state components from voltage surge.

3.3.5.6 PLC. A PLC controller shall be provided for the salt bath with the following minimum components and features shall determine equivalency are as follows:

- a. Honeywell HC900 Programmable Logic Controller (PLC). The system software shall be off-the-shelf, user friendly, and commercially available.
- b. MiniTrend digital chart recorder. The unit shall be capable of exporting filed in PDF or MS Excel format onto stand-alone computer with a storage capacity up to one Terabyte (1TB).
- c. The human interface shall have industrial-rated touch screen(s) with integral protection from liquid and or physical damage.
- d. The system shall provide all the necessary inputs and outputs (I/O) to monitor and control all required alarms, buzzers, fans, motors, pumps, valves, gas systems, sensors, and other devices necessary for safe and proper operation of the system

3.3.5.7 Burner Control, Modulation, and Operating Modes. The Contractor shall program the gas control system to allow smooth transition between pilot/manual modes and automatic mode for Burner Number 1 and Burner Number 2 independently. Each burner shall have individual proportional control of combustion air and natural gas flow to achieve the required combustion ratio and stable temperature control. Burner operation, modulation, and safety interlocks and operating conditions shall be controlled and monitored by the main furnace PLC.

3.3.6 Motors. Motors shall be rated for continuous duty. Alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG 1 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of the proposal,

the vendor shall indicate which document they will utilize for compliance to the requirements. Each motor shall have an identification plate in accordance with paragraph 3.6 to identify manufacturer, model number, serial number, voltage, amperage, horsepower, phase, frequency, and frame number. Motor horsepower shall be the manufacturer's standard or standard optional horsepower as indicated on the manufacturer's design and construction drawing(s).

3.3.7 Facility Connections. Electrical connections made to the facility electrical system shall conform to NFPA 70.

3.3.8 Hour Meter. The equipment shall be equipped with a non-resetting type, hour meter to display accumulated operating time of the drive motor(s), mounted on the main electrical panel visible from the outside, sealed to prevent the entrance of dust and moisture, and shall be mounted to withstand shock and vibration generated by the equipment. The meter shall have a range of 0 to 99,999.9 hours in increments of one (1) tenth of one (1) hour.

3.4 Size and Capacity. The size and capacity of the machine shall be not less than specified in Table I.

Table 1: Equipment Specifications and Requirements

Salt Bath Specification	Requirement
Salt Bath Dimensions (L, W, H)	153" X 66" X 112"
Maximum Temperature	1100°F
Minimum Temperature	480°F
Operating Temperature	850°F – 950°F
Temperature Uniformity	± 5°F throughout qualified work zone, IAW SAE AMS 2750, Class 1
Temperature Accuracy	± 2°F of setpoint, IAW SAE AMS 2750
Heat Up Rate to Operating Temperature	Ambient to operating temperature within 24 hours (from solid to liquid)
Quench Tank Specification	Requirement
Quench Tank Dimensions (L, W, H)	153" X 66" X 112"
Tank Material	Stainless Steel

3.5 Accuracy Tolerances. Sensor calibration accuracy shall be in accordance with Table 1 SAE AMS 2750.

Output of sensors shall be converted to temperature readings by instruments specified in SAE AMS 2750 or instruments of equal or greater accuracy. Instruments shall be calibrated in accordance with Table 7 SAE AMS 2750 and shall be traceable to NIST or other internationally recognized standards organization.

Resolution requirements for analog recording instruments shall be in accordance with Table 8 SAE AMS 2750.

Process recording instrument print and chart speeds shall be in accordance with Table 9

SAE AMS 2750.

Digital recording instruments shall be in accordance with SAE AMS 2750.

3.6 Marking on Instruments, Control Panels, and Plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.6.1 Lubrication Plate. A lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information.

- Points of Lubricant Application
- Servicing Interval
- Type of Lubricant
- Viscosity
- Military or Federal Specification for Each Lubricant (If available)
- Size and Type of Each Lubricant Filter

3.6.2 Nameplate. A nameplate shall be permanently and securely attached to each machine. The nameplate data shall meet the requirements of NFPA 79 and contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine, and a suffix identified in the manufacturer's permanent records.

- Nomenclature
- CAGE Code
- Manufacturer's Name
- Manufacturer's Model Designation
- Manufacturer's Serial Number
- Power Input (Volts, Total Amps, Phase, Frequency)
- Contract Number
- Date of Manufacture
- Full-load current for each supply
- Ampere rating of the largest motor or load
- Short-circuit current rating of the industrial control panel
- Max ampere rating of short-circuit and ground fault protective device, where provided
- Electrical diagram number(s) or number of the index to the electrical drawings

3.6.3 Identification Marking of Military Property. An Item Unique Identification Marking (IUID) shall be provided in accordance with MIL-STD-130N and all applicable documents within the standard with Machine Readable Information (MRI) for item identification marking and automatic data capture. The application of Human Readable Information (HRI) shall be used in combination with MRI and free text. At a minimum this tag shall contain the information listed in paragraph 3.6.2 Nameplate.

3.7 Technical Data. Three (3) hard copies and one (1) electronic copy of technical data shall be provided with the equipment. The electronic copy shall be provided via Department of Defense (DoD) Secure Access File Exchange (SAFE), secure email, or alternative means. Technical data shall contain specifications, instructions for the operation, maintenance of the equipment and recommended critical spare parts list. All copies of technical data shall be legible and written in the English language and in PDF (.pdf) format.

3.7.1 Drawings and Deliverables. The contractor shall provide complete design and installation drawings for the equipment and system, including electrical control schematics and power distribution schematics. Drawings shall be prepared in accordance with ASME Y14.100 and shall conform to applicable industry drafting standards.

3.7.1.1 Pre-Installation Drawings. Design and installation drawings shall be submitted for Government review and approval prior to fabrication and installation. No fabrication or installation shall proceed until the drawings have been reviewed and approved.

3.7.1.2 Format Requirements. All drawings shall be provided in both hard copy and digital formats. Hard copies shall be D-size (24 in. x 36 in.). Digital files shall be provided in AutoCAD (.dwg) format.

3.7.1.3 Redlines and As-Built Drawings. The contractor shall maintain redline drawings during installation to document all field changes. Upon completion of installation, the contractor shall submit final as-built drawings reflecting the actual installed configuration of the system. Final as-built drawings shall be provided in both D-size hard copy and AutoCAD digital format.

3.7.1.4 Drawing Quality and Use. All drawings shall be of sufficient detail and quality to support operation, maintenance, troubleshooting, and future modifications of the system.

3.7.2 Plan of Action and Milestones (POA&M). The contractor shall provide a POA&M to identify the contractor's performance dates of all major events from contract award to completion of all contract requirements. The initial POA&M shall be delivered thirty (30) calendar days After Contract Award (ACA). The milestones shall be prepared using a recognized project management technique (preferably Microsoft Project). The Government and the contractor shall review the report on a monthly basis. The contractor shall notify the Government prior to making any changes to the POA&M. All changes to the POA&M shall be approved in advance by the Government. The POA&M shall detail such items as engineering, procurement of major hardware, assembly, operational debug, factory acceptance, shipment, installation, training, and final acceptance test.

3.7.3 Acceptance Test Plan (ATP). The Contractor shall prepare an Acceptance Test Plan (ATP). The ATP shall be a step-by-step written instruction on how to verify compliance during Contract Quality Assurance Inspection and the Final Inspection/Acceptance Test specified in Sections 4.4 and 4.5. The Manufacturer's standard testing procedures shall be included in the ATP. Any procedures necessary for compliance with Section 4, which are not covered by the Manufacturer's standard testing procedures, shall be identified and

included in the ATP. The initial ATP shall be submitted thirty (30) calendar days ACA for Government review and approval. The accepted test procedures shall become part of this contract.

3.8 Installation. Contractor shall be responsible for the attached Installation Responsibilities Sheet and Local Documents. The contractor shall follow the customers' hours of operation; 6:00am to 2:30pm, Monday through Friday excluding all national holidays, unless informed otherwise.

3.8.1 Salt Removal and Disposal. The Contractor shall drain/pump, remove, and containerize existing salt solution from salt bath tank. The Contractor shall coordinate with FRCSW POC to manage proper disposal of existing salt solution and residues prior to the Contractor removing inner tank lining for refurbishment.

3.8.2 Water Removal and Disposal. The Government shall drain, remove, containerize, and manage disposal of existing water and any remaining residues.

3.8.3 Removal of Existing Quench Tank. The Contractor shall be responsible for the complete removal of the existing water quench tank that is being replaced by the new quench tank. The existing tank shall be removed and delivered to the FRCSW DRMO lot for proper disposal.

3.9 Training. After satisfactory completion of acceptance testing of the system, the services of a qualified representative(s), who is proficient in the English language, shall be provided for specialized training to familiarize Government personnel with the equipment and to help ensure reliable performance and maximum service life, during normal usage. The training shall be for ten (10) FRCSW personnel including operators, maintenance, and engineers. Training shall be performed at the Government facility and shall not be less than five (5) consecutive eight (8) hour workdays, between 6:00 am and 2:30 pm excluding weekends and federal holidays. The course shall utilize as-built drawings and manuals provided under this contract and shall include both classroom and hands-on portions. All printed training aids shall be supplied for the course, be in the English language, and become the property of the Government. The following items shall be covered during the training:

- a. Operating the furnace from start to end of heat-treat cycle.
- b. Programming the furnace, loading material in furnace basket, heat-treating material in furnace, and using upgraded tank operation to quench and rinse material.

3.9.1 Familiarization. The Contractor shall familiarize Government personnel with the equipment, including a description of each major system and subsystem, a description of all safety and emergency features, and familiarization with the related technical data, manuals, and documentation provided under the contract.

3.9.2 Demonstration. The Contractor shall demonstrate to Government personnel all normal operations, safety functions, and emergency procedures associated with the equipment, including startup, normal operation, shutdown, and response to alarms and fault conditions.

3.9.3 Hands-On Operation. The Contractor shall observe Government personnel during hands-on operation of the equipment and shall provide assistance, instruction, and corrective guidance as needed to ensure personnel are capable of safely and correctly operating the system.

4.0 QUALITY ASSURANCE.

4.1 Responsibility for Inspection. The contractor shall be responsible for the performance of all inspection requirements specified herein and in accordance with the attached Quality Assurance Provision (QAP). The Government reserves the right to witness any inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. The Contractor shall be responsible for ensuring all items meet the requirements of sections 3 and 4.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

- a. Final Inspection/Acceptance test (Destination) (see 4.4).

4.3 Inspection Conditions. All inspections, tests, and examinations shall be performed in an indoor facility with ambient temperature conditions in the range of 55° F to 100° F and 10% to 95% relative humidity.

4.4 Final Inspection/Acceptance Test (Destination). Final Inspection and acceptance test shall be performed on the machine to ensure conformance with this purchase description. The acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.5 and all tests in 4.6. The machine shall pass the examination and all tests to be accepted.

4.5 Examination. The equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification.

4.6 Tests. Unless otherwise specified within this document, all programs, instruments, materials, material handling equipment, qualified personnel, and tools required to perform and evaluate the tests shall be furnished by the contractor. All measurement tools and indicating devices shall be calibrated in accordance with the attached QAP. If a malfunction occurs during the test, the test shall be restarted each time until all tests in 4.7 are completed without a malfunction. Three (3) consecutive failures without completion of the test may be considered cause for rejection of the system. For the purpose of this test, a “failure” is defined as any equipment malfunction which requires remedial action to restore the system to full operation in accordance with this specification.

4.6.1 Operational Test. The Contractor shall perform an operational demonstration of the molten salt bath system to verify the system is fully functional, safely operable, and ready

for production use. This operational test shall not replace or supersede the SAT or TUS, but shall demonstrate overall system operability, safety interlocks and alarms, control stability, and temperature monitoring and recording capability. The demonstration shall be conducted within the normal operating range of 850°F to 950°F and shall include verification of safety devices and interlocks, controlled heat-up, stable operation at temperature, and representative load disturbance and recovery. The Contractor shall verify continuous temperature recording with correct channel identification and time/date stamping and demonstrate the ability to retrieve and export recorded data. Upon completion, the Contractor shall perform a normal shutdown of each system and verify transition to a safe condition. The Contractor shall provide a brief operational test report documenting test conditions, recorded trends, alarm and interlock checks performed, and any deficiencies identified. Successful completion of this operational test shall demonstrate the equipment is operationally ready for production use, subject to completion and acceptance of SAT and TUS requirements where applicable.

4.6.2 Performance Test. The equipment shall be tested in accordance with the agreed upon ATP referenced in paragraph 3.7.3. The equipment shall be tested in accordance with the applicable requirements of NFPA 86, Standard for Ovens and Furnaces, and SAE AMS 2750.

4.6.3 Temperature Uniformity Survey (TUS) and System Accuracy Test (SAT). The TUS and SAT shall comply with SAE AMS 2750, Pyrometry Section 3.3.1 for a Class 1 furnace. Contractor shall perform TUS and SAT at customer's location. Detailed step-by-step instructions shall be provided to perform the TUS, SAT and to calibrate temperature instrumentation. Any ancillary equipment, fixtures etc. required to perform the TUS and SAT will be provided with the furnace.

4.6.4 Audible Noise Level Test. The equipment shall be tested for compliance with paragraph 3.1.4. "Audible Noise Level".

INSTALLATION RESPONSIBILITIES

Government

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Contractor

- ☐ a. Provide molten salt bath as “core” to be upgraded.
- ☐ b. Provide and install replacement quench tank.
- ☒ c. Furnish labor and material handling equipment for off-loading and placing item on foundation.
- ☒ c. Provide and install anchor bolts and nuts.
- ☒ d. Set and rough level the machine on its foundation
- ☒ e. Level and align equipment.
- ☒ f. Connect equipment to the provided utilities hook-up.
- ☒ g. Provide all necessary tools, gages, and instrumentation necessary to perform the required tests.
- ☒ h. Provide and charge all systems with fluids in accordance with manufacturer's instructions.
- ☐ i. Material for performance testing at Government facility.
- ☐ k. Material handling, Rigging equipment and Qualified Personnel for performance testing.
- ☒ j. Testing for lead and asbestos.
- ☐ k. Abatement of lead and asbestos.

GOVERNMENT RESPONSIBILITIES

- ☒ Provide utilities hook-up within 20 feet, line of sight of the machine within the same compartment.